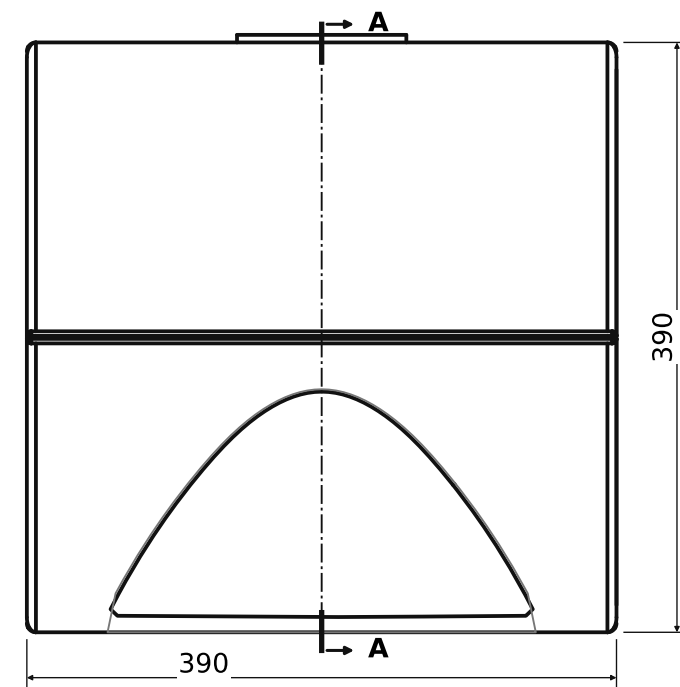
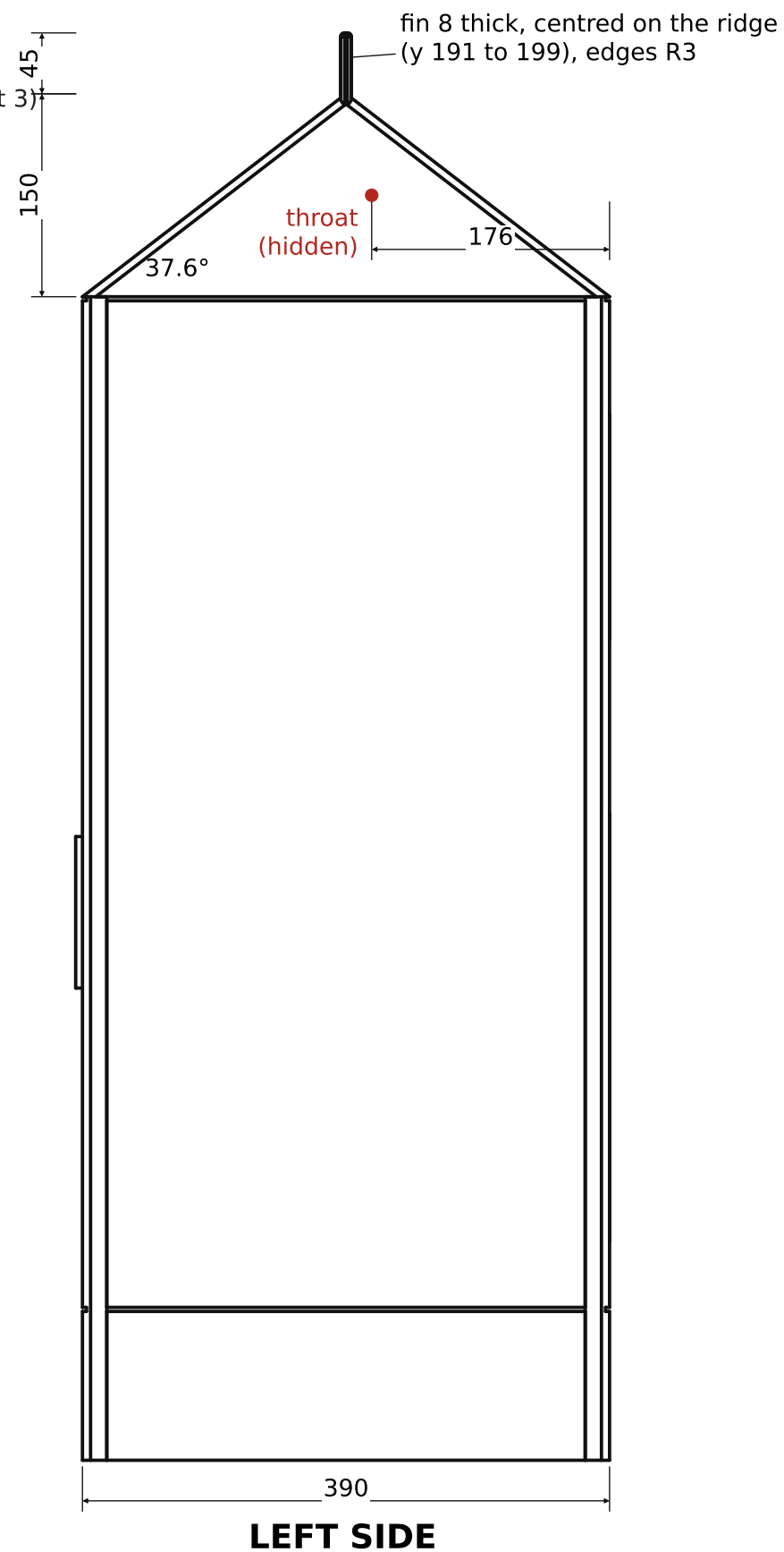
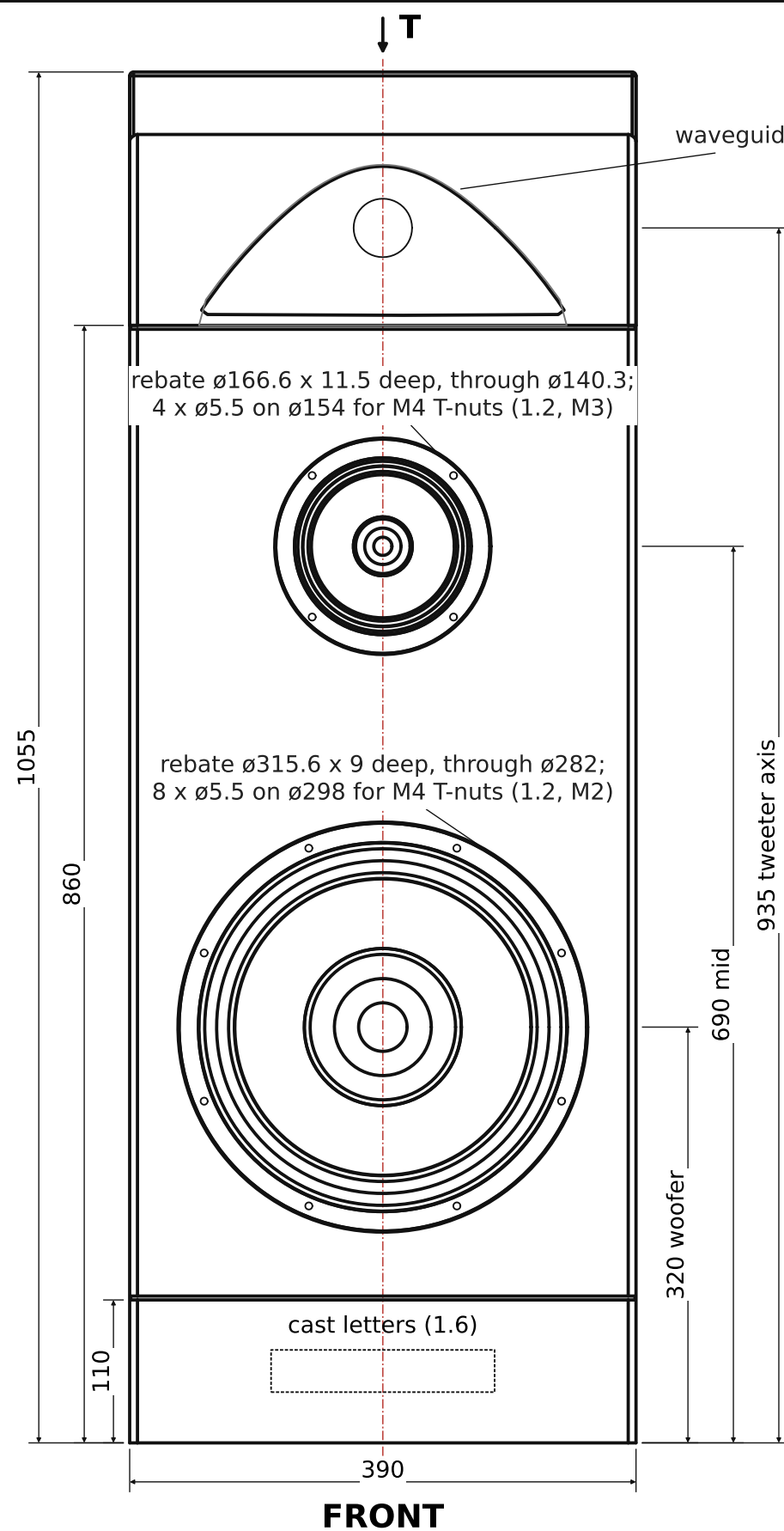




Notes: notes 1.1 to 1.6, sheet 7

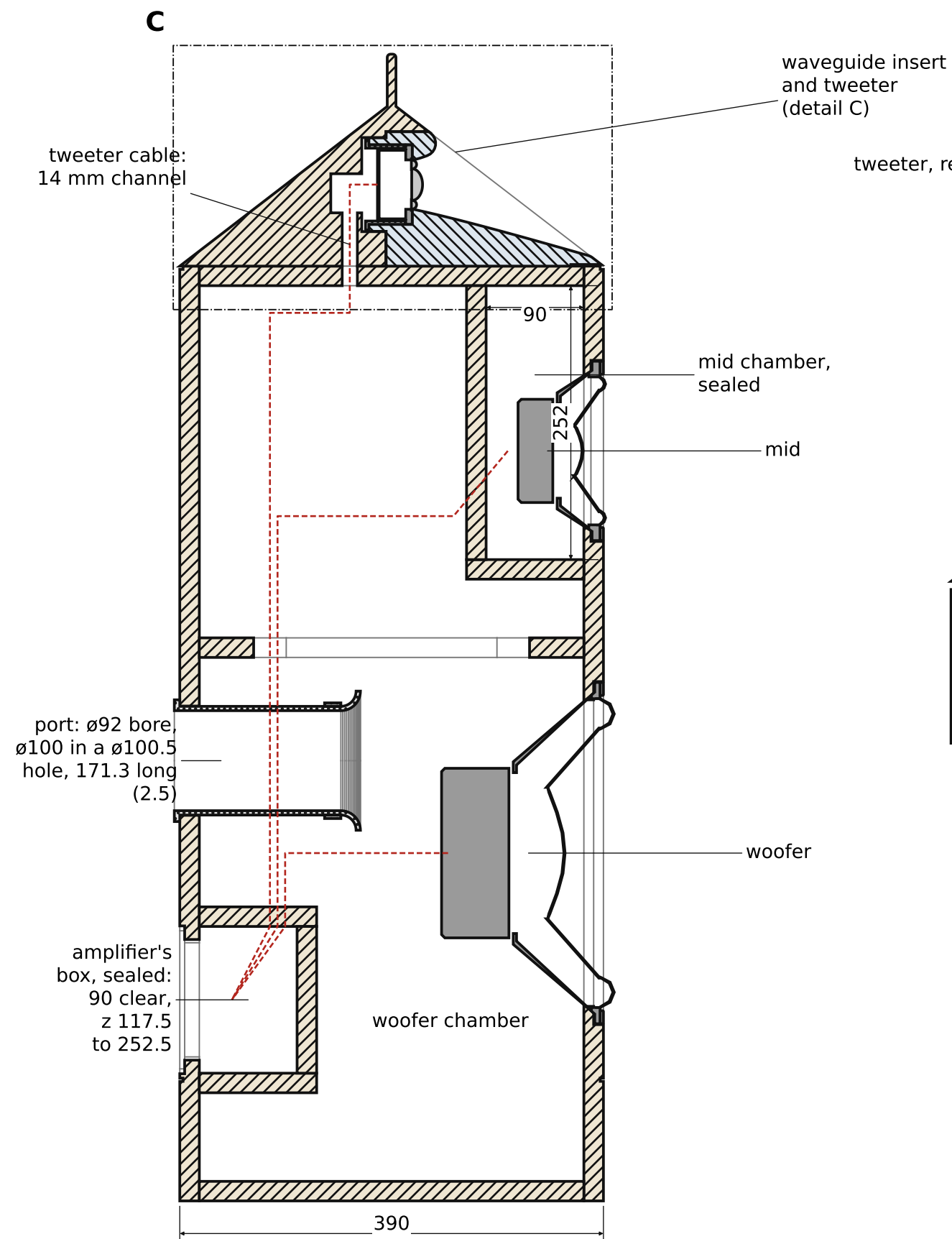
General arrangement

earmilk floorstander. Front, left side and top at 1:5. Heights from the floor; depths from the front face.



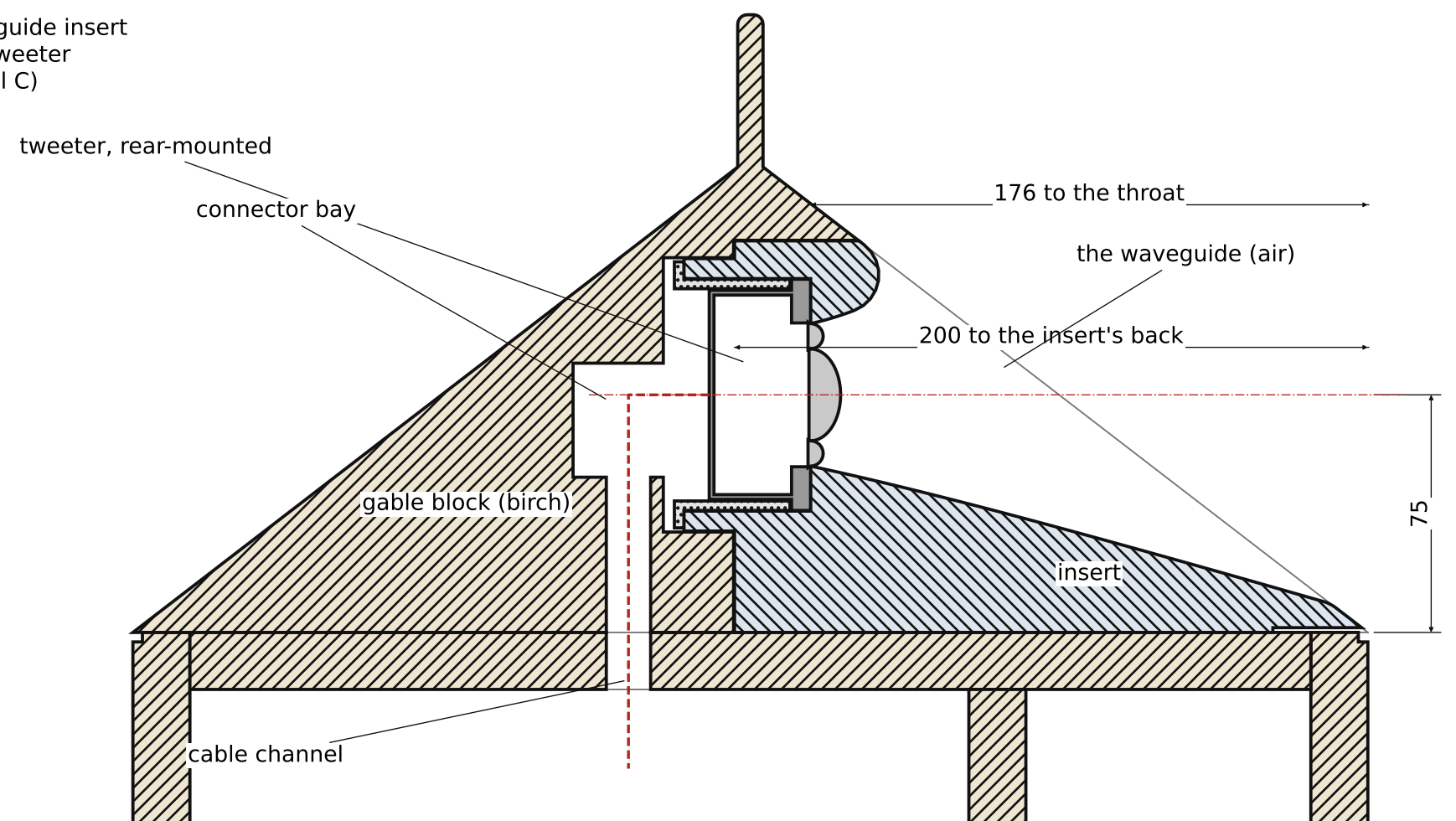
mm | issue 2026-10-09 | from fab/sheets.py (the CAD)
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

1 of 8
EM-FS-001 rev D



SECTION A-A

red dashed: the cables' runs to the amplifier (sheet 4)

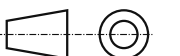


DETAIL C, 1:2

The roof: notes 2.1 to 2.5, sheet 7

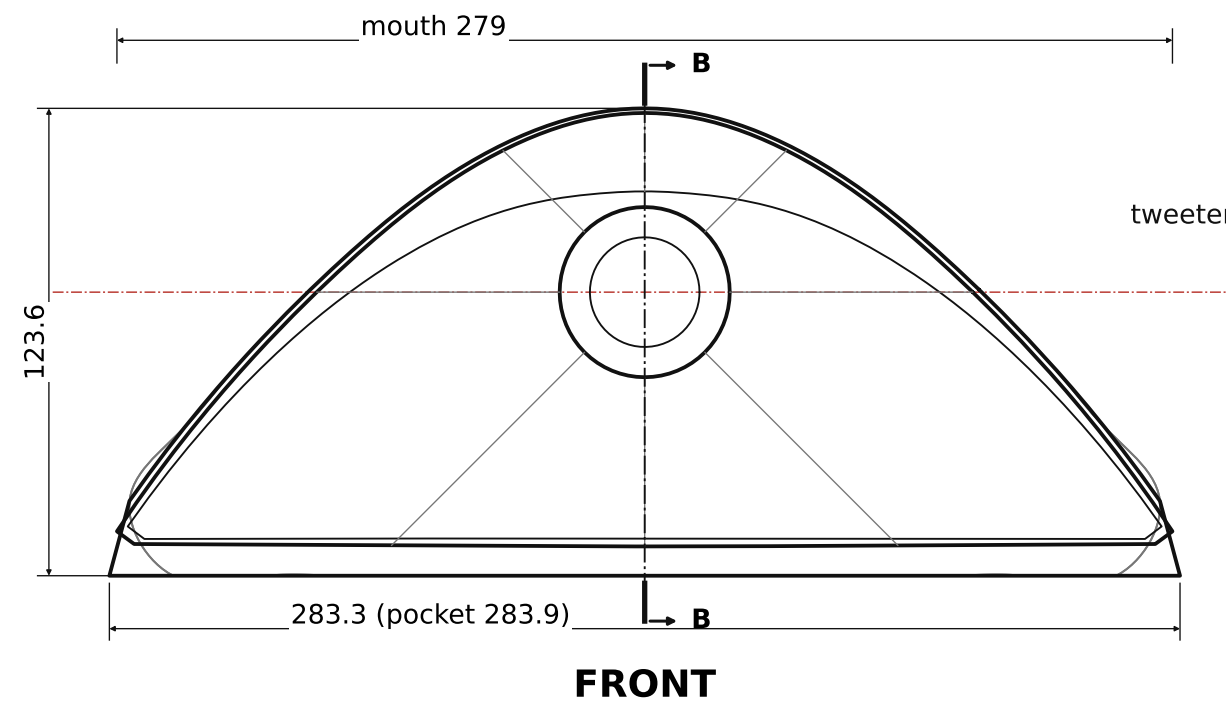
Section on the centreline

earmilk floorstander. Section A-A on the centre plane (x 195), from the left, at 1:5; the roof at 1:2. The insert hatched the other way.

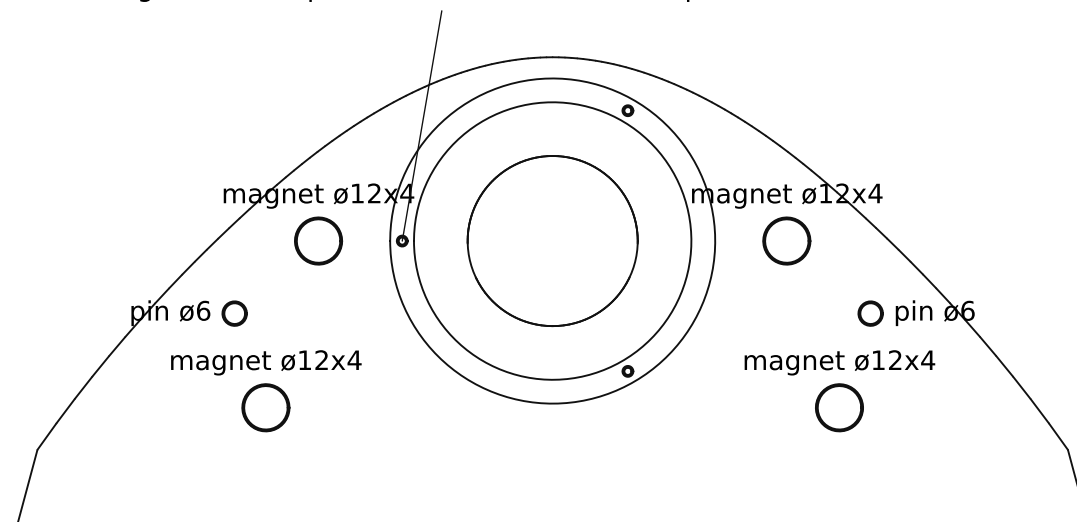


mm | issue 2026-10-09 | from fab/sheets.py (the CAD)
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

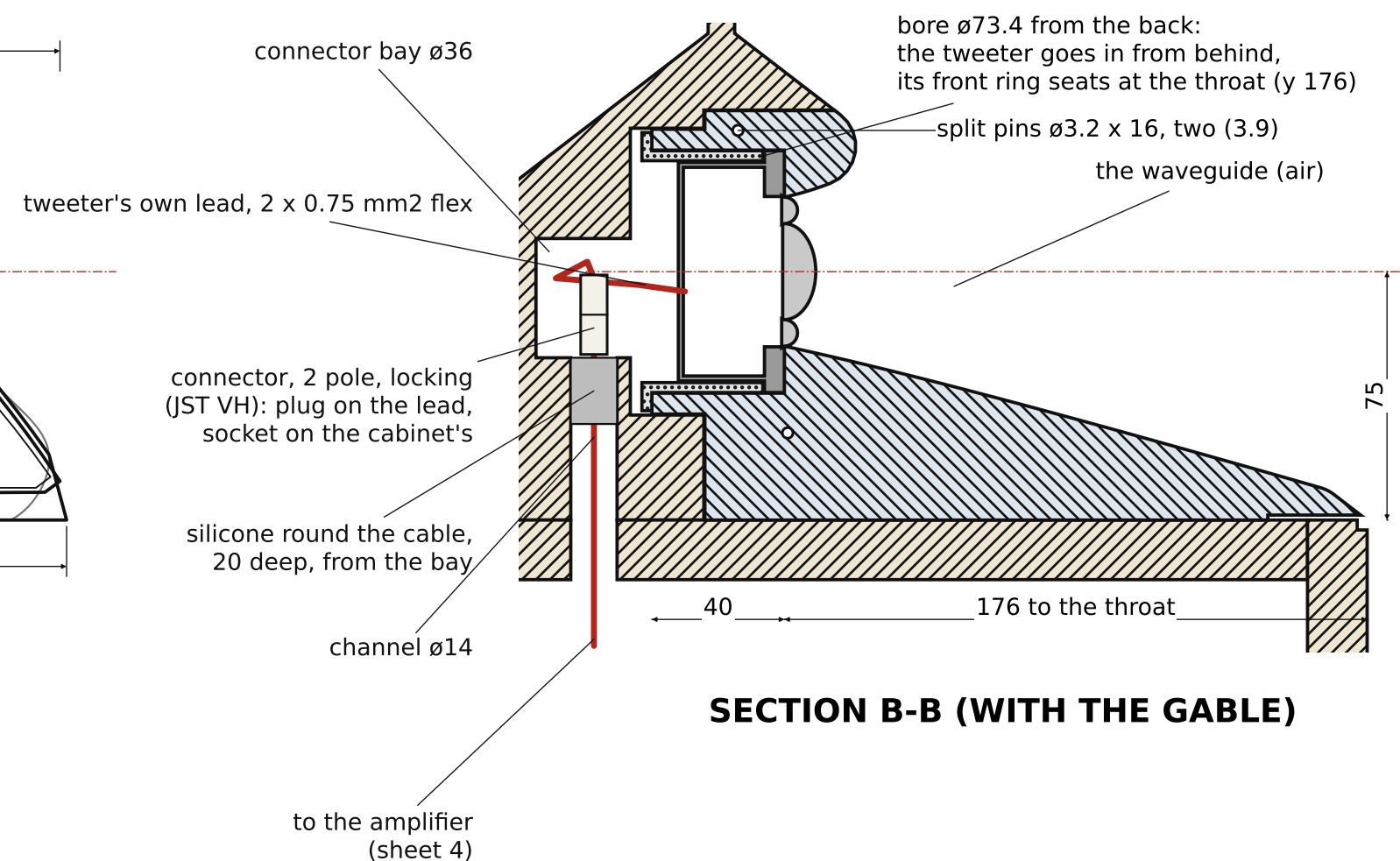
2 of 8
EM-FS-002 rev D



3 x $\varnothing 2.4 \times 10$ pilots on $\varnothing 79.6$ in the boss's back face, one on the horizontal at the speaker's right (left here), 120° apart, for the retaining sleeve's 3.0 x 12 thread-forming screws for plastics (WN 1411 / PT K30, pan head)



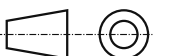
BACK (THE FACE THAT MEETS THE POCKET)



Fitting the tweeter and the insert: notes 3.1 to 3.9, sheet 7

Waveguide insert and tweeter mount

earmilk floorstander. The insert at 1:2: front, section on the axis, back; the pocket; fitting and service.

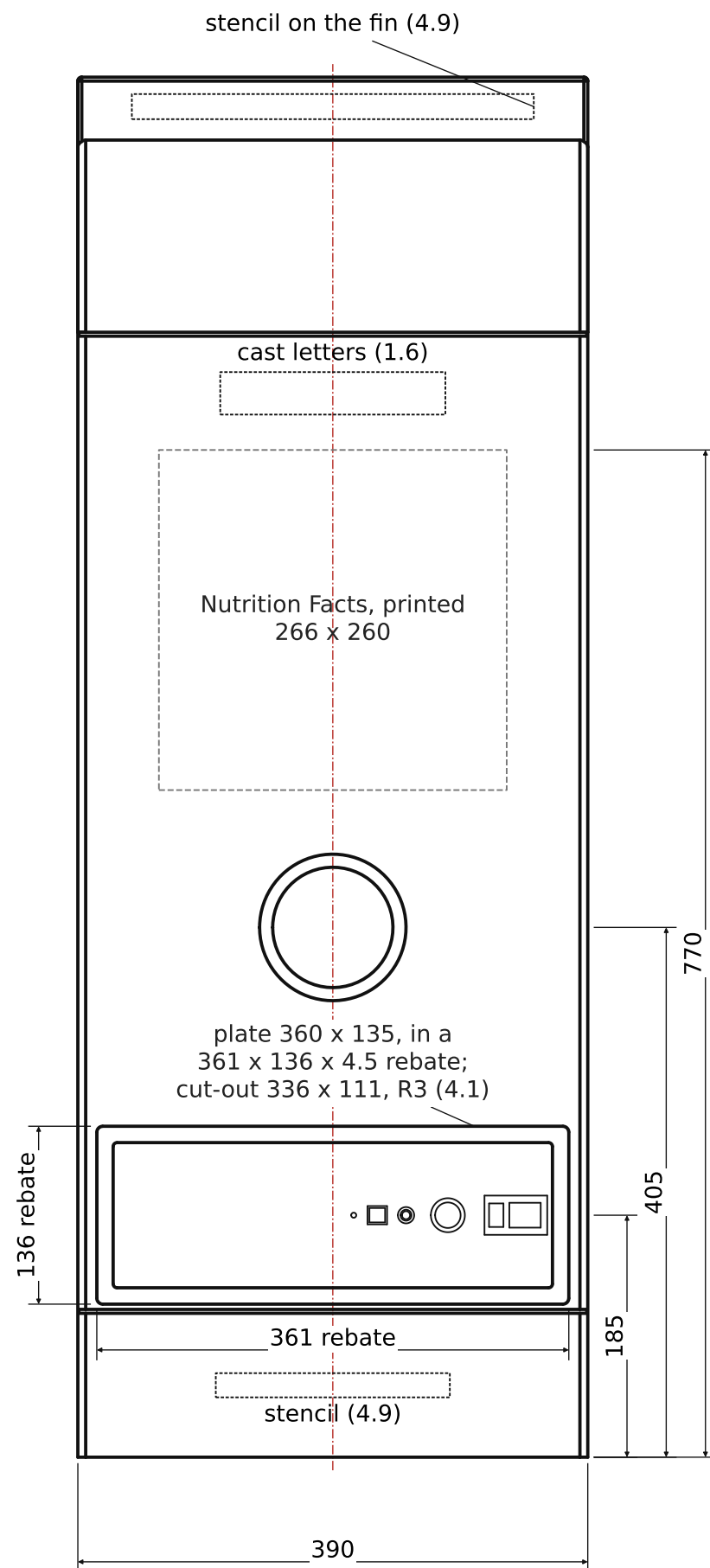


mm | issue 2026-10-09 | from fab/sheets.py (the CAD)

Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

3 of 8

EM-FS-003 rev D

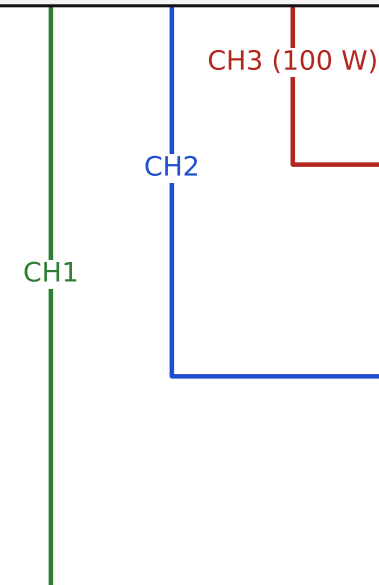


BACK

Mains
IEC C14 inlet, switch

Signal in
XLR or RCA
USB for the DSP

Hypex FusionAmp FA253
3 channels, DSP crossover and EQ (ADAU1452)
250 + 250 W into 4 ohm, 100 W tweeter channel
on its side across the back's foot
in its own sealed box (not airtight itself)
filters: Hypex Filter Design, over USB
mains in on the plate (IEC C14 with switch)



Tweeter, SB Satori TW29DN-B
29 mm dome, faceplate off, on the insert
1 mm2 pair, red +, black -

Midrange, SB Satori MR16P-8
6.5 in papyrus, its own sealed chamber
1.5 mm2 pair, red +, black -

Woofer, Dayton Audio RSS315HF-4
12 in, 4 ohm, vented to 32 Hz
2.5 mm2 pair, red +, black -

Runs, connectors and seals: notes 4.1 to 4.9, sheet 8

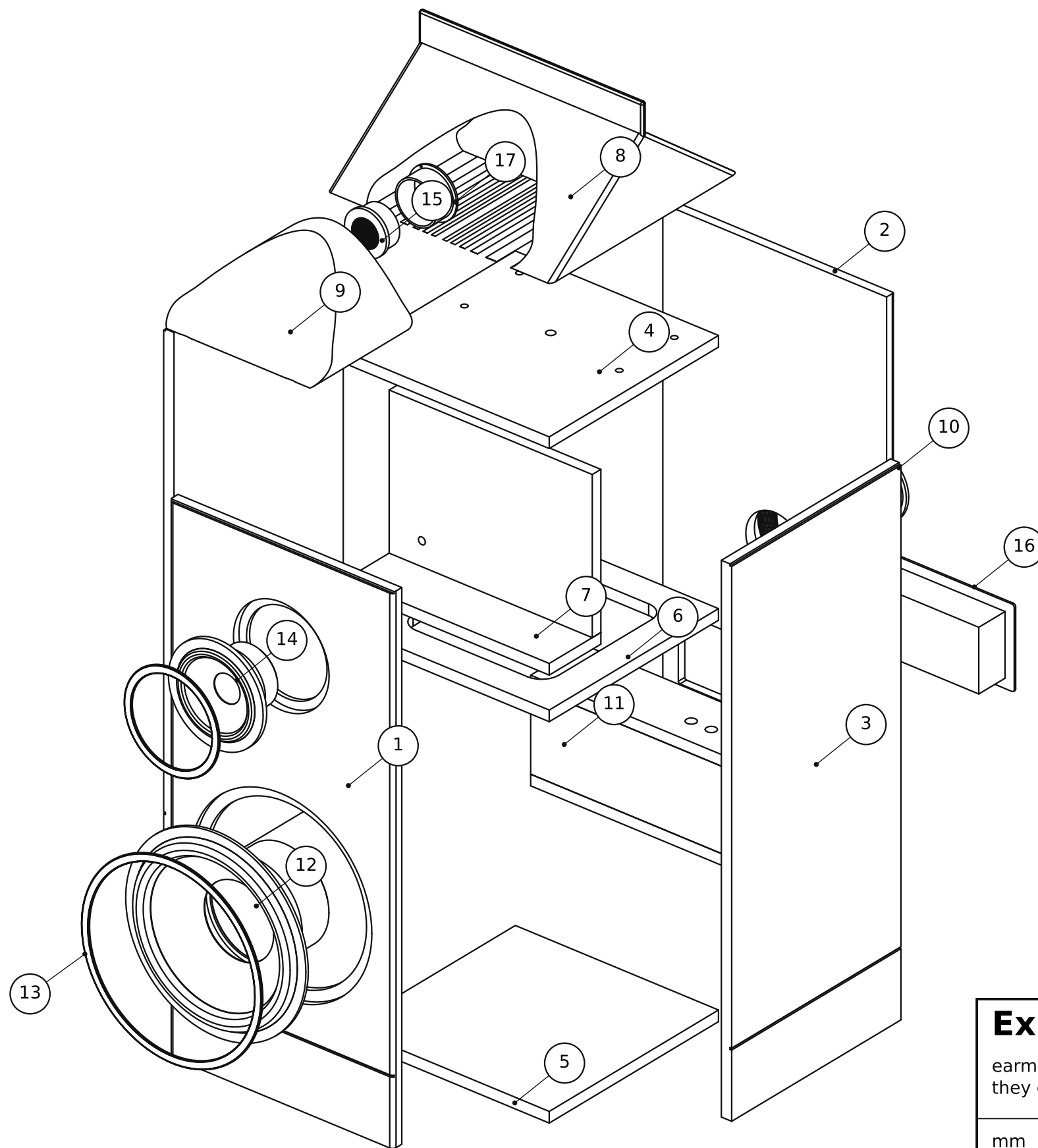
Back and wiring

earmilk floorstander. The back at 1:5 and the amplifier, drivers, cables and connectors. Active: the amplifier's DSP is the crossover.



mm | issue 2026-10-09 | from fab/sheets.py (the CAD)
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

4 of 8
EM-FS-004 rev D



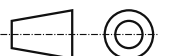
Parts

- ① **Front baffle x1**
18 mm Baltic birch, CNC: cut-outs and rebates
- ② **Back x1**
18 mm birch: port hole, amplifier cut-out, rebate
- ③ **Sides x2**
18 mm birch
- ④ **Top panel x1**
18 mm birch: the 14 mm cable hole, 4 dowel holes
- ⑤ **Bottom panel x1**
18 mm birch
- ⑥ **Window brace x1**
18 mm birch
- ⑦ **Mid shelf and divider x2**
18 mm birch; the divider's 12 mm cable hole
- ⑧ **Gable block x1**
laminated birch, CNC, or printed in six pieces
- ⑨ **Waveguide insert x1**
SLA tough resin or MJF nylon; magnets, pins
- ⑩ **Port x1**
printed tube and flare collar, 92 bore
- ⑪ **Amplifier box x3**
birch floor, lid (3 glands) and front; sealed
- ⑫ **Woofer x1**
Dayton Audio RSS315HF-4, 12 in, 4 ohm
- ⑬ **Trim rings x2**
printed, satin black, over each frame, screws
- ⑭ **Midrange x1**
SB Acoustics Satori MR16P-8, 6.5 in
- ⑮ **Tweeter x1**
SB Acoustics Satori TW29DN-B, faceplate off
- ⑯ **Amplifier x1**
Hypex FusionAmp FA253, DSP, 3 channels
- ⑰ **Tweeter retainer x1**
printed sleeve over the motor, 3 screws

Glue-up order: G1 to G7, sheet 8

Exploded view and parts

earmilk floorstander. Isometric, not to scale: the parts pulled apart along the way they go together, and what each one is.

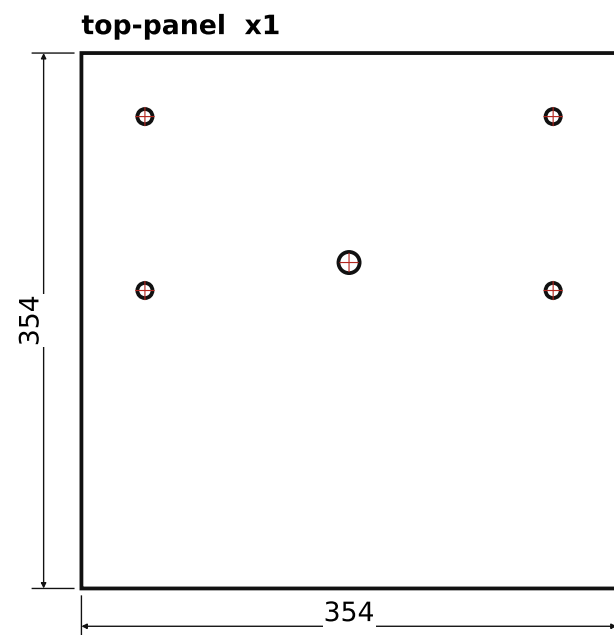


mm | issue 2026-10-09 | from fab/sheets.py (the CAD)

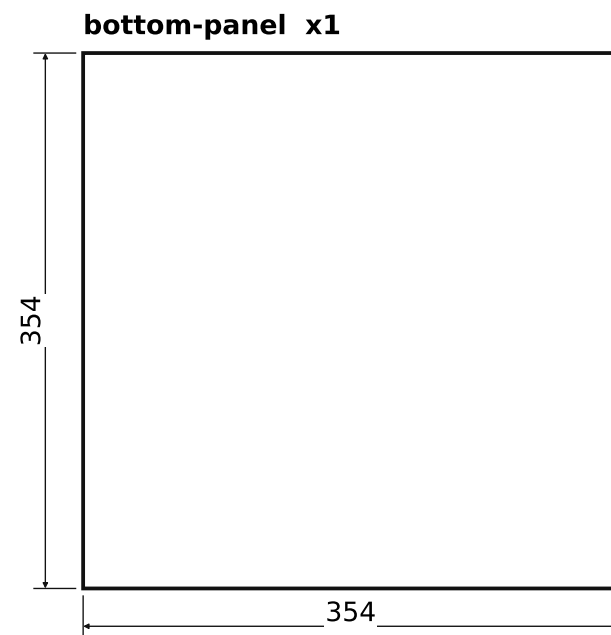
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

5 of 8

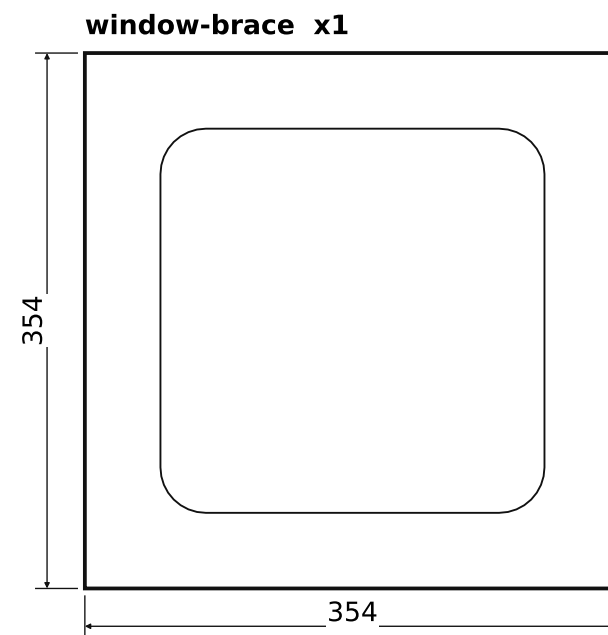
EM-FS-005 rev D



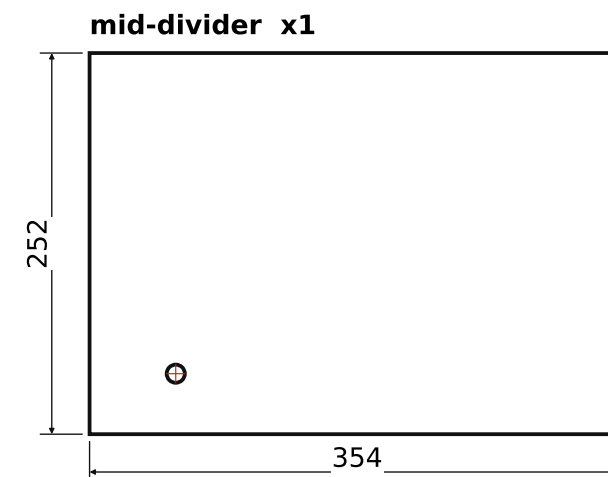
ø14 at (177, 215.5): cut inside
4 x ø10 at (42, 197), (312, 197), (42, 312),
(312, 312): drill ø10, 10 deep
upper face up (front edge at the bottom of the
drawing); dowels register the gable block; the
tweeter cable's hole is sealed with silicone
from the bay after wiring



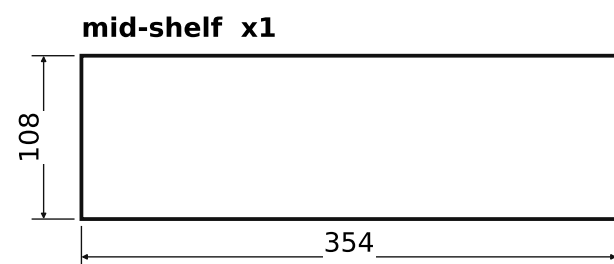
either face



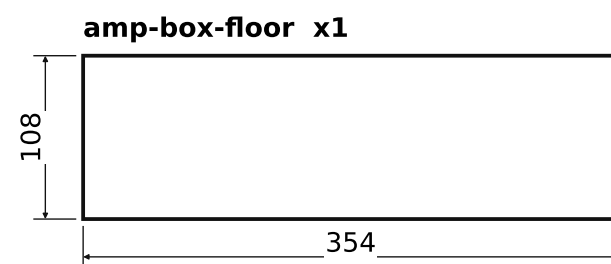
cut inside: 50 to 304 across, 50 to 304 up, R30
corners
sits at z 500 to 518, glued to all four walls



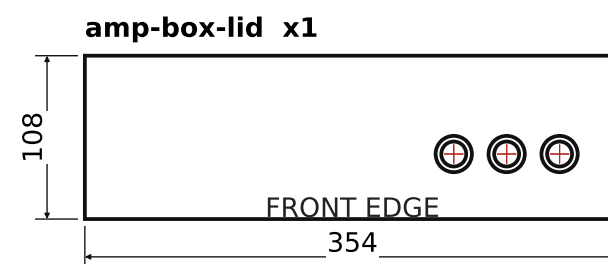
ø12 at (57, 40): cut inside
y 108 to 126, z 590 to 842; seal the wire hole
after wiring



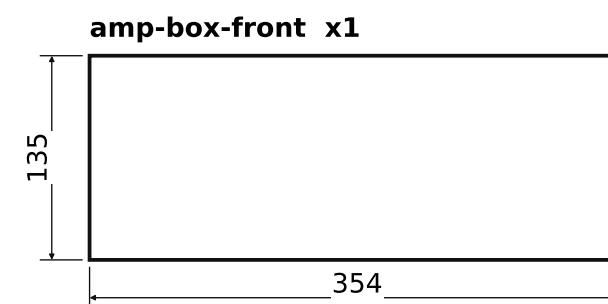
z 572 to 590, from the baffle back to under the
divider



the amplifier box's floor, z 99.5 to 117.5,
against the back



3 x ø16.5 at (314, 43), (279, 43), (244, 43):
cut inside
3 x ø24 at (314, 43), (279, 43), (244, 43):
pocket 8 mm deep from the underside, concentric
with each gland's hole (cut the holes, turn the
panel over, centre on them; before the lid goes
in)
the amplifier box's lid, z 252.5 to 270.5, its
front edge (the bottom of the drawing) over the
box's front; 3 ø16.5 holes for M16 x 1.5
glands, one per cable, 314, 279, 244 from the
left, 43 from the front edge, each nut in a ø24
x 8 counterbore from the underside



the amplifier box's front, between floor and
lid, 90 in front of the back's inner face; glue
and seal all round

Trim rings, section at 2:1 (stl/trim-ring-*.stl): flat, flush with the finish

woofer: ø314 / ø290 x 3



channel open to the bore to ø307.0, 2 deep

mid: ø165 / ø141 x 3



channel from ø145.0 to ø163.0, 2 deep

the channel clears the M4 x 20 heads (F1): 8 across, 1.6 high or less

Panel details

earmilk floorstander. At 1:5: each panel seen from its outer (upper) face, from its
lower-left corner as its DXF; holes from that corner. 18 mm birch.



mm | issue 2026-10-09 | from fab/sheets.py (the CAD)

Tolerances: panels ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

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EM-FS-006 rev D

Measure first

- M1** Ply: 18 thick; the sides and inner panels are 390 - 2 x 18 = 354 wide. For a real thickness t: 390 - 2t (heights, pockets and rebates unchanged); set WALL = t in params.py and run fab/build.sh again.
- M2** Woofer (Dayton RSS315HF-4): flange 5 thick sets the rebate, 9 deep; the frame's diameter + 1.6 sets the rebate's ø315.6; the cut-out ø282; 8 holes on ø298; the surround at its glue line + 2 sets the trim ring's bore, ø290.
- M3** Mid (SB Satori MR16P-8): flange 7.5 thick sets the rebate, 11.5 deep; the frame's diameter + 1.6 sets the rebate's ø166.6; the cut-out ø140.3; 4 holes on ø154; the surround at its glue line + 2 sets the trim ring's bore, ø141.
- M4** SB Acoustics Satori TW29DN-B: the front ring ø73 sets the bore, ø73.4; the motor ø66 sets the sleeve's bore, ø67; the dome and surround, 45 across, set the throat. Print nothing until it is measured.
- M5** Hypex FusionAmp FA253: the plate 360 x 135 sets the rebate, 361 x 136; the module sets the cut-out, 336 x 111; the screw holes come from the plate in hand.

Sheet 1: Notes

- 1.1** Cabinet: 18 mm Baltic birch, glued. The vertical corners and the gable's hips are rounded 6 mm, the fin's edges 3 mm. 3 x 3 mm shadow lines at z 110 and z 857.
- 1.2** The drivers sit flush: each rebate is the measured flange + 1.0 (a 1.5 mm closed-cell foam gasket, pressed) + 3 (the ring): woofer 9 for a 5 flange, mid 11.5 for a 7.5 flange; measure each flange before cutting (M2, M3): the rebate is the flange + 4.0 only while 7.5 of birch stays under it for the T-nuts. The screws and T-nuts as F1, pressed in before the glue-up. A printed trim ring over each flange (stl/trim-ring-*.stl, woofer ø314 / ø290 x 3, mid ø165 / ø141 x 3, flat, a channel in its back over the screws' heads: the sections at 2:1 on sheet 6) held by three 5 mm dots of neutral-cure silicone, 0.8 reveal all round. Its inner ø is the surround at its glue line + 2: measure the drivers and print the rings last (the woofer's ring needs a 320 bed, or a print service).
- 1.3** The tweeter is at the throat of a waveguide insert in the roof (sheet 3): axis z 935, 176 behind the front face. The shape is set by simulation (fab/out/acoustics/waveguide).
- 1.4** The back (port, amplifier, label) is drawn on sheet 4 with the wiring.
- 1.5** Finish: colour coat and 2K clear; the Facts printed between them.
- 1.6** The cast letters (out/metal/wordmark-letters.*), 1.5 proud: 172.2 x 32.5 of ink, centred across the plinth's front, the badge's centre z 55; on the back the same at z 813. Place and drill them from the 1:1 templates, out/metal/wordmark-template-front.pdf and -back.pdf, after the clear coat.

Sheet 2: The roof

- 2.1** The waveguide insert (light blue) sits in a pocket in the gable block, 0.3 mm clear all round: the pocket +0.2/0, the insert 0/-0.15, so 0.3 to 0.65 a side. Paint only the insert's face: mask its sides, base and back and the pocket's walls. It slides out forward, level, with the tweeter on it.

- 2.2** The tweeter is rear-mounted: it goes in from behind through a ø73.4 bore and its front ring seats on the ring round the throat, its dome and surround filling the 45 mm throat, so the wall runs on from the surround with no step. A printed sleeve over its motor holds it there, screwed to the boss's back face (sheet 3).
- 2.3** Its lead leaves the boss's open back into the connector bay behind it (ø36 x 35 deep), where it plugs into the cabinet's lead; that runs down a 14 mm channel through the block and the top panel, sealed round the cable with neutral-cure silicone from the bay.
- 2.4** Four magnets and two pins in the insert's back face meet their partners in the pocket's back wall (sheet 3).
- 2.5** The port: a 92 bore, its tube 100 outside in a ø100.5 hole through the back, centre z 405; its ø112 x 5 flange glued to the finish. 153.3 from the flange's face, with the 18 flare collar 171.3 long. Printed 163.3 and trimmed at its inner end until the impedance dip sits at the tuning (README).

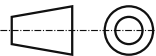
Sheet 3: Fitting the tweeter and the insert

- 3.1** Solder the tweeter's own lead (150 mm of 2 x 0.75 mm2 silicone-insulated flex, twisted (about 2.2 mm each)) to its tabs and crimp the plug half of a two-pole locking connector (JST VH) on its end.
- 3.2** Pass the tweeter in from behind through the ø73.4 bore, dome first, and seat its front ring on a 0.5 mm foam gasket on the throat's seat. Slide the printed retaining sleeve (stl/tweeter-retainer.stl: bore ø67, outside ø73, flange ø84 x 3) over its motor, lead through it, and drive its 3 3.0 x 12 thread-forming screws for plastics (WN 1411 / PT K30, pan head) into the boss until the ring is held: no thread to SB's own screw holes is needed. Measure the motor first: the sleeve's bore is its diameter + 1.
- 3.3** Glue the magnets into the insert's back with epoxy, polarity marked, and their partners into the pocket's back wall, opposite poles out. Bond the two pins into the insert with epoxy.
- 3.4** The cabinet's lead runs up the channel and ends in the socket standing in the bay, 97 mm above the top panel's underside (30 past the bay's floor); seal the channel round it with 20 mm of neutral-cure silicone from the bay before the first fitting. The slack is the tweeter's own lead, 150 mm of 2 x 0.75 mm2 silicone-insulated flex, twisted (about 2.2 mm each): hold the insert just in front of its pocket, plug the lead into the socket in the bay, and slide the insert home; the lead folds into the sleeve and the bay.

- 3.5** Slide the insert in, level, until the pins seat and the magnets pull it home: its face flush with the roof, the seam even. Fit: the pocket +0.2/0, the insert 0/-0.15 (0.3 to 0.65 a side); paint only the insert's face.
- 3.6** Service: pull it out by a ribbon loop glued in the 12 x 1.5 x 30 groove under its front edge, unplug, and the tweeter comes out with it.
- 3.7** The pocket (gable-block.step): the insert's outline + 0.3 a side (283.9 wide at its base), from the slope to its back wall at y 200.3; the boss's bore ø86.6 to y 222.5; the connector bay ø36 from y 216 to 251 on an axis at z 927; the channel ø14 at x 195, y 233.5 down to the top panel; internal corners R3 or less (the insert's match); dowel holes ø10 x 20 in the base at (60, 215), (330, 215), (60, 330), (330, 330).
- 3.8** Fixings, from the centreline and the insert's base (mirrored in the pocket's wall): magnets ±75.9 across, 30.9 up; ±61.9 across, 75.0 up; pins ±84.1 across, 55.8 up. Magnets ø12 x 4 in ø12.2 x 4.3 holes, epoxied, poles marked; pins ø6 x 10 bonded with epoxy in ø6.1 x 5.5 in the insert, a slip fit in ø6.2 x 5.5 in the wall (these holes are outside the title block's +0.2/-0).
- 3.9** Printing the insert: one piece from a service (SLA tough resin or MJF nylon, a build of 290 x 220 x 130 or more) is the default, with no seam in the waveguide. On a desktop resin printer (218 x 123 x 220) print it in halves split at the centre plane, the section B-B: each half fits only tilted (about 209 x 117 x 213). Join them with two ø3.2 x 16 steel pins across the split, at y 175.0, z 886.3 and y 190.0, z 977.7 (3.2 holes, 8 deep each side), and epoxy; fill and sand the seam on the waveguide's wall flush.

Notes: measure first; sheets 1 to 3

earmilk floorstander. What to measure before cutting or printing, and the notes sheets 1 to 3 point to by number.



mm | issue 2026-10-09 | from fab/sheets.py (the CAD)
Tolerances: panels ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Sheet 4: Runs, connectors and seals

- 4.1
- The amplifier's plate (360 x 135 x 3) lies on 3 mm EPDM tape in its 4.5 deep rebate, R4.5 corners, its module through the 336 x 111 cut-out (R3 corners: a 6 mm cutter or smaller). Drill its screw holes ø3.5 through the 13.5 left under the rebate, from the plate in hand, centred in its flange, 8 or 10 to suit it: 4.3 x 16 self-tapping pan heads (Hypex's 4.3 x 25 kit also works, its tips 7 mm into the sealed box).
- 4.2
- Woofers (CH1): from its gland in the amplifier box's lid straight to the woofer's terminals, 0.6 m of 2.5 mm². Fit 6.3 mm push-on terminals to suit the driver.
- 4.3
- Mid (CH2): from its gland up through the brace's window, 30 mm in from its back edge, then through the 12 mm hole in the mid chamber's divider (x 75, z 630), 1.0 m of 1.5 mm². Seal the hole round the cable with putty: the mid's chamber must stay closed.
- 4.4
- Tweeter (CH3): from its gland up through the brace's window to the top panel, through the 14 mm channel in the top panel and the gable block, 1.2 m of 1.0 mm², ending 97 above the top panel's underside in the socket of a two-pole locking connector (JST VH) standing in the bay behind the insert. The supplied 1.25 m Hypex harness is too short once it is spliced: extend it (4.6).
- 4.5
- Seal the tweeter cable in its channel with 20 mm of neutral-cure silicone from the bay (sheet 3). The insert's pocket is open to the room through its 0.3 mm seam, so an unsealed channel would be a leak in the woofer's box.
- 4.6
- Round-sheathed cable (H05VV-F 2 x 2.5, 2 x 1.5 and 2 x 1.0 mm², about 9.0, 8.0 and 6.4 across), so each gland grips its cable and the silicone seals round the tweeter's; splice the tweeter's extension to Hypex's harness inside the amplifier's box (crimped butt splices under heat-shrink).
- 4.7
- The amplifier box's lid: 3 M16 x 1.5 long-thread (15 mm) cable glands, one per cable (clamping about 4.5 to 10), in ø16.5 holes 314, 279, 244 from the left side's inner face and 43 from the lid's front edge, each lock nut in a ø24 x 8 counterbore under the lid. Fit them after the finish, the bodies from above and the nuts from below through the amplifier's cut-out; feed the cables, then tighten. The box itself is glued and sealed; its front is the woofer chamber's wall.
- 4.8
- Polarity, delay and the crossover (about 2.8 kHz tweeter to mid, from the waveguide study) are set in the DSP: wire red to + throughout and let the filters do the rest.
- 4.9
- The stencils (out/marks/stencil*.svg), sprayed through vinyl masks before the clear: SHAKE WELL, 26 mm type, centred on the plinth's back at z 55; OPEN OTHER SIDE, 27 mm type with its arrow, centred on the fin's back face.

Glue-up, in order

- G1
- Lay the back face down. Glue both sides and the bottom to it.
- G2
- The amplifier box: its floor, front and lid between the sides, against the back; seal its joints.
- G3
- The window brace at z 500, glued to the sides and the back.
- G4
- The mid chamber's shelf, then its divider.

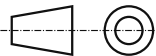
- G5
- Press the T-nuts into the front baffle's inside face, then glue the baffle on.
- G6
- The top panel last, inside all four walls.
- G7
- Clamp across the sides at every inner panel. Then the gable block on its dowels.

Fixings and the pocket's floor

- F1
- Drivers: M4 x 20 low-profile (wafer) button head, A2, the head 1.6 high or less and 8 across or less (ISO 7380 is 2.2 high: the trim ring's channel is 2 deep). M4 T-nuts, flange 15 or less across (woofer), 12 or less across (mid), barrel 8 or less (woofer), 6 or less (mid), pressed in before the glue-up.
- F2
- Before the gable block goes on, sand the front panel's top edge flush with the top panel to 0.1: the insert sits on both. Mask the pocket's floor with the insert's sides, base and back when you spray: paint only the insert's face.

Notes: sheets 4 to 6; glue-up and fixings

earmilk floorstander. The notes sheets 4 to 6 point to by number; the order of the glue-up; the fixings.



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Tolerances: panels ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15